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LIGHT SOURCE AND SUBSTRATE

Abstract

A light source includes a substrate, a light-emitting device, and a first bonding member. The first bonding member is disposed between a first electrode of the light-emitting device and a first upper surface of a first metal member of the substrate to bond the first electrode and the first metal member, and is disposed between a second electrode of the light-emitting device and a second upper surface of a second metal member of the substrate to bond the second electrode and the second metal member. An upper portion of the insulating member of the substrate is located between the first electrode and the second electrode of the light-emitting device. Each of the first lower surface of the first metal member, the second lower surface of the second metal member, and the lower portion of the insulating member is exposed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2024-035042, filed Mar. 7, 2024, the contents of which is hereby incorporated by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present invention relates to a light source and a substrate.

2. Description of Related Art

[0003] Japanese Patent Publication No. 2022-102279 describes a configuration in which a protruding member having insulating properties is provided between a first electrode and a second electrode of an LED chip.

SUMMARY

[0004] Embodiments of the present invention can provide a light source and a substrate that can improve heat dissipation.

[0005] According to one aspect of the present invention, a light source includes a substrate, a light-emitting device disposed on the substrate, and a first bonding member disposed between the substrate and the light-emitting device. The substrate includes a first metal member having a first upper surface and a first lower surface located on a side opposite to the first upper surface, a second metal member having a second upper surface and a second lower surface located on a side opposite to the second upper surface, the second metal member being spaced apart from the first metal member in a first direction, and an insulating member including a lower portion located between the first metal member and the second metal member in the first direction, and an upper portion extending, from the lower portion, to be higher than the first upper surface and the second upper surface. The light-emitting device includes a semiconductor layered body having a first element surface facing the substrate and a second element surface located on a side opposite to the first element surface, a first electrode disposed at the first element surface, and a second electrode disposed at the first element surface to be spaced apart from the first electrode in the first direction. The first bonding member is disposed between the first electrode and the first upper surface of the first metal member and bonds the first electrode and the first metal member and is disposed between the second electrode and the second upper surface of the second metal member and bonds the second electrode and the second metal member. The upper portion of the insulating member is located, in the first direction, between the first electrode and the second electrode of the light-emitting device. Each of the first lower surface of the first metal member, the second lower surface of the second metal member, and the lower portion of the insulating member is exposed.

[0006] According to another aspect of the present invention, a substrate includes a first metal member having a first upper surface and a first lower surface located on a side opposite to the first upper surface, a second metal member having a second upper surface and a second lower surface located on a side opposite to the second upper surface, the second metal member being spaced apart from the first metal member in a first direction, and an insulating member including a lower portion located between the first metal member and the second metal member in the first direction, and an upper portion extending, from the lower portion, to be higher than the first upper surface and the second upper surface. A width, in the first direction, of the lower portion of the insulating member is greater than a width, in the first direction, of the upper portion of the insulating member. The lower portion of the insulating member has a first lower lateral surface facing an inner lateral

surface of the first metal member and a second lower lateral surface facing an inner lateral surface of the second metal member. The substrate further includes a bonding member bonding the inner lateral surface of the first metal member and the first lower lateral surface and bonding the inner lateral surface of the second metal member and the second lower lateral surface, the bonding member being located at an upper surface of the lower portion of the insulating member.

[0007] According to an aspect of the present invention, a light source and a substrate that can improve heat dissipation can be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] A more complete appreciation of embodiments of the invention and many of the attendant advantages thereof will be readily obtained by reference to the following detailed description when considered in connection with the accompanying drawings.

[0009] FIG. 1 is a schematic plan view illustrating a light source according to an embodiment.

[0010] FIG. 2 is a schematic cross-sectional view taken along a line II-II in FIG. 1.

[0011] FIG. 3 is a schematic cross-sectional view illustrating a light source according to a variation of the embodiment.

[0012] FIG. 4 is a schematic plan view illustrating a step of a method for manufacturing a substrate according to an embodiment.

[0013] FIG. 5 is a schematic cross-sectional view taken along a line V-V in FIG. 4.

[0014] FIG. 6 is a schematic cross-sectional view illustrating a step of the method for manufacturing the substrate according to the embodiment.

[0015] FIG. 7 is a schematic cross-sectional view illustrating a step of the method for manufacturing the substrate according to the embodiment.

[0016] FIG. 8 is a schematic cross-sectional view illustrating a step of the method for manufacturing the substrate according to the embodiment.

[0017] FIG. 9 is a schematic cross-sectional view illustrating a step of the method for manufacturing the substrate according to the embodiment.

[0018] FIG. 10 is a schematic cross-sectional view illustrating a step of the method for manufacturing the substrate according to the embodiment.

[0019] FIG. 11 is a schematic cross-sectional view illustrating a step of the method for manufacturing the substrate according to the embodiment.

DETAILED DESCRIPTION OF EMBODIMENT

[0020] Embodiments of the present invention will be described below with reference to the drawings. The following embodiments are merely examples for embodying the technical concept of the present invention, and the present invention is not limited to the following embodiments. Further, dimensions, materials, shapes, relative arrangements, or the like of constituent members described in the embodiments are not intended to limit the scope of the present disclosure thereto, unless otherwise specified, and are merely exemplary. Note that the size, positional relationship, or the like of members illustrated in each of the drawings may be exaggerated for clarity of description. Further, in the following description, members having the same terms and reference characters represent the same members or members of the same quality, and a detailed description of these members will be omitted as appropriate. As a cross-sectional view, an end view illustrating only a cut surface may be illustrated.

[0021] In the following description, terms indicating specific directions or positions (e.g., “upper”, “lower”, and other terms including those terms) may be used. However, these terms are used merely to make it easy to understand relative directions or positions in the referenced drawing. As long as the relative direction or position is the same as that described in the referenced drawing

using the term such as “upper” or “lower”, in drawings other than the drawings of the present disclosure, actual products, and the like, components need not be arranged in the same manner as that in the referenced drawing. For example, on the assumption that there are two members, the positional relationship expressed as “upper (or lower)” in the present specification may include a case in which the two members are in contact with each other and a case in which the two members are not in contact with each other and one of the two members is located above (or below) the other of the two members. Further, in the present specification, unless otherwise specified, a case in which a member covers an object to be covered includes a case in which the member is in contact with the object to be covered and directly covers the object to be covered, and a case in which the member is not in contact with the object to be covered and indirectly covers the object to be covered. Further, in the present specification, a width in a specific direction represents a maximum width in the specific direction.

[0022] In the drawings described below, a direction along an X-axis is referred to as a first direction X, and the first direction X indicates a predetermined direction in a plane parallel to a light-emitting surface of a light source according to an embodiment.

[0023] A direction along a Y-axis is referred to as a second direction Y, and the second direction Y indicates a direction orthogonal to the first direction X in the plane parallel to the light-emitting surface. A direction along a Z-axis is referred to as a third direction Z, and the third direction Z indicates a direction orthogonal to the above-described light-emitting surface.

Light Source

[0024] A light source **1** according to a first embodiment will be described with reference to FIGS. **1** and **2**. The light source **1** includes a substrate **10**, a light-emitting device **30** disposed on the substrate **10**, and a first bonding member **41** disposed between the substrate **10** and the light-emitting device **30**.

Light-Emitting Device

[0025] The light-emitting device **30** includes a semiconductor layered body **33**, a first electrode **31**, and a second electrode **32**. The light-emitting device **30** is, for example, a light-emitting diode (LED). Alternatively, the light-emitting device **30** can be a laser diode (LD).

[0026] The semiconductor layered body **33** has a first element surface **33A** facing the substrate **10**, and a second element surface **33B** located on a side opposite to the first element surface **33A** in the third direction Z.

[0027] The semiconductor layered body **33** includes a nitride semiconductor. In the present specification, it is assumed that examples of the nitride semiconductor include semiconductors having all compositions of a chemical formula expressed by $\text{In}_{\text{sub.x}}\text{Al}_{\text{sub.y}}\text{Ga}_{\text{sub.1-x-y}}\text{N}$ ($0 \leq x \leq 1$, $0 \leq y \leq 1$, $x+y \leq 1$) in which the composition ratios of x and y are changed within the respective ranges. Further, it is assumed that examples of the nitride semiconductor also include a semiconductor further containing a group V element other than nitrogen (N) in the chemical formula described above, and a semiconductor further containing, in the chemical formula described above, various elements added to control various physical properties such as the conductivity type of the semiconductor. The semiconductor layered body **33** includes an n-side semiconductor layer, a p-side semiconductor layer, and an active layer located between the n-side semiconductor layer and the p-side semiconductor layer. The active layer is a light-emitting layer that emits light and has a multiple quantum well (MQW) structure including a plurality of barrier layers and a plurality of well layers, for example. Light emitted by the active layer is visible light or ultraviolet light, for example.

[0028] The first electrode **31** is disposed at the first element surface **33A**. The second electrode **32** is disposed at the first element surface **33A** so as to be spaced apart from the first electrode **31** in the first direction X. The first electrode **31** is a cathode electrode in the light-emitting device **30**, and is electrically connected to the n-side semiconductor layer in the semiconductor layered body **33**. The second electrode **32** is an anode electrode in the light-emitting device **30**, and is electrically

connected to the p-side semiconductor layer in the semiconductor layered body **33**. As a material of the first electrode **31** and the second electrode **32**, gold, platinum, silver, aluminum, nickel, titanium, or the like can be used, for example.

[0029] The light-emitting device **30** can include an element substrate such as a sapphire substrate, a gallium nitride substrate, a silicon substrate, or the like, on the second element surface **33B** of the semiconductor layered body **33**. For example, when the element substrate is a sapphire substrate, the element substrate can be removed using a laser lift-off method, and the light-emitting device **30** may not include the element substrate.

Substrate

[0030] The substrate **10** includes a first metal member **11**, a second metal member **12**, and an insulating member **20**. The substrate **10** supports the light-emitting device **30**. Further, the substrate **10** is also a wiring member that electrically connects the light-emitting device **30** to an external circuit.

[0031] The first metal member **11** has a first upper surface **11A** and a first lower surface **11B**. The first upper surface **11A** faces the first electrode **31** of the light-emitting device **30** in the third direction Z. The first lower surface **11B** is located on a side opposite to the first upper surface **11A** in the third direction Z. Further, the first metal member **11** has an inner lateral surface **11C** and an outer lateral surface **11D** located on a side opposite to the inner lateral surface **11C** in the first direction X.

[0032] The second metal member **12** has a second upper surface **12A** and a second lower surface **12B**. The second upper surface **12A** faces the second electrode **32** of the light-emitting device **30** in the third direction Z. The second lower surface **12B** is located on a side opposite to the second upper surface **12A** in the third direction Z. Further, the second metal member **12** has an inner lateral surface **12C** facing the inner lateral surface **11C** of the first metal member **11** in the first direction X, and an outer lateral surface **12D** located on a side opposite to the inner lateral surface **12C** in the first direction X.

[0033] The first metal member **11** can include a plating **13** facing the first electrode **31** in the third direction Z, and the surface of the plating **13** can be the first upper surface **11A**. The first metal member **11** can include a plating **15** located on a side opposite to the plating **13** in the third direction Z, and the surface of the plating **15** can be the first lower surface **11B**. The second metal member **12** can include a plating **14** facing the second electrode **32** in the third direction Z, and the surface of the plating **14** can be the second upper surface **12A**. The second metal member **12** can include a plating **16** located on a side opposite to the plating **14** in the third direction Z, and the surface of the plating **16** can be the second lower surface **12B**. When the first metal member **11** and the second metal member **12** include the plating, a first bonding member **41** described below is less likely to diffuse into the first metal member **11** and the second metal member **12**, and generation of voids in the first bonding member **41** can be reduced. The platings **13**, **14**, **15**, and **16** contain, for example, at least one metal selected from nickel (Ni), palladium (Pd), titanium (Ti), platinum (Pt), and gold (Au). Among them, nickel is effective in suppressing the diffusion of the first bonding member **41**. When the first bonding member **41** does not easily wet and spread with respect to nickel, palladium or gold can be further formed on nickel in order to make the first bonding member **41** easily wet and spread.

[0034] As illustrated in FIG. 1, the first metal member **11** and the second metal member **12** are spaced apart from each other in the first direction X. The first metal member **11** and the second metal member **12** extend in the second direction Y. The first metal member **11** and the second metal member **12** contain, for example, at least one metal selected from copper (Cu) and aluminum (Al).

[0035] The insulating member **20** includes a lower portion **22** and an upper portion **21**. The upper portion **21** is located on the lower portion **22** in the third direction Z. The lower portion **22** and the upper portion **21** extend in the second direction Y. The insulating member **20** is made of, for example, a ceramic. The insulating member **20** is, for example, a ceramic sintered compact in

which the lower portion **22** and the upper portion **21** are integrally provided. The insulating member **20** includes, for example, silicon nitride, aluminum oxide, aluminum nitride, silicon carbide, or silicon oxide. Alternatively, the insulating member **20** can be made of a metal covered with a ceramic, glass, or a resin.

[0036] The lower portion **22** of the insulating member **20** is located between the first metal member **11** and the second metal member **12** in the first direction X. The lower portion **22** has a first lower lateral surface **22A** facing the inner lateral surface **11C** of the first metal member **11**, and a second lower lateral surface **22B** facing the inner lateral surface **12C** of the second metal member **12**.

[0037] The upper portion **21** extends from the lower portion **22** in the third direction Z, so as to be higher than the first upper surface **11A** of the first metal member **11** and the second upper surface **12A** of the second metal member **12**. The upper portion **21** has an upper surface **21C** facing the first element surface **33A** of the semiconductor layered body **33**. A part, on the upper surface **21C** side, of the upper portion **21** is located between the first electrode **31** and the second electrode **32** of the light-emitting device **30** in the first direction X. The upper portion **21** has a first upper lateral surface **21A** facing an inner lateral surface of the first electrode **31**, and a second upper lateral surface **21B** facing an inner lateral surface of the second electrode **32**.

[0038] The lower portion **22** has a lower surface **22D** located on a side opposite to the upper surface **21C** of the upper portion **21** in the third direction Z. Each of the first lower surface **11B** of the first metal member **11**, the second lower surface **12B** of the second metal member **12**, and the lower surface **22D** of the lower portion **22** of the insulating member **20** is exposed.

[0039] The light source **1** can be disposed on a mounting substrate. The first lower surface **11B** of the first metal member **11**, the second lower surface **12B** of the second metal member **12**, and the lower surface **22D** of the lower portion **22** of the insulating member **20** face a mounting surface of the mounting substrate. The first lower surface **11B** of the first metal member **11** and the second lower surface **12B** of the second metal member **12** are bonded to a wiring portion of the mounting substrate by a bonding member such as solder. Accordingly, the first metal member **11** and the second metal member **12** are electrically connected to the wiring portion of the mounting substrate.

[0040] The width, in the first direction X, of the upper portion **21** of the insulating member **20** disposed between the first electrode **31** and the second electrode **32** of the light-emitting device **30** is smaller than the width, in the first direction X, between the first electrode **31** and the second electrode **32**. The width, in the first direction X, of the lower portion **22** of the insulating member **20** is greater than the width, in the first direction X, of the upper portion **21** of the insulating member **20**. Accordingly, the spacing distance, in the first direction X, between the first lower surface **11B** of the first metal member **11** and the second lower surface **12B** of the second metal member **12** can be made greater than the distance between the electrodes of the light-emitting device **30**, and thus the first lower surface **11B** and the second lower surface **12B** become less likely to short-circuit on the mounting substrate due to a bonding member.

First Bonding Member

[0041] The first bonding member **41** has conductivity. As the first bonding member **41**, solder can be used, for example. The solder includes, for example, a gold-tin alloy. Alternatively, as the first bonding member **41**, a mixture of a resin and metal particles can be used. As the metal particles, for example, silver particles can be used.

[0042] The first bonding member **41** is disposed between the first electrode **31** of the light-emitting device **30** and the first upper surface **11A** of the first metal member **11**, and bonds the first electrode **31** and the first metal member **11**. When the first metal member **11** includes the plating **13**, the first bonding member **41** is bonded to the plating **13**. The first electrode **31** is electrically connected to the first metal member **11** via the first bonding member **41**.

[0043] The first bonding member **41** is disposed between the second electrode **32** of the light-emitting device **30** and the second upper surface **12A** of the second metal member **12**, and bonds the second electrode **32** and the second metal member **12**. When the second metal member **12**

includes the plating **14**, the first bonding member **41** is bonded to the plating **14**. The second electrode **32** is electrically connected to the second metal member **12** via the first bonding member **41**.

[0044] The semiconductor layered body **33** including the active layer in the light-emitting device **30** generates heat when it emits light. The heat generated by the semiconductor layered body **33** is dissipated to the first metal member **11** via the first electrode **31** and the first bonding member **41**, and is dissipated to the second metal member **12** via the second electrode **32** and the first bonding member **41**.

[0045] Furthermore, according to the present embodiment, the heat generated by the semiconductor layered body **33** can be transferred to the insulating member **20**. The heat transferred from the semiconductor layered body **33** to the insulating member **20** can be transferred to the first electrode **31**, the second electrode **32**, the first metal member **11**, and the second metal member **12**, which are made of a metal material having a higher thermal conductivity than that of the insulating member **20**. Further, when the lower surface **22D** of the lower portion **22** is in contact with the mounting surface of the mounting substrate, heat can be directly transferred from the insulating member **20** to the mounting substrate.

[0046] An inter-electrode region of the first element surface **33A** located between the first electrode **31** and the second electrode **32** of the semiconductor layered body **33** is a region in which heat cannot be dissipated to the substrate **10** via the first electrode **31** and the second electrode **32** and heat is more likely to be concentrated than in regions of the first element surface **33A** in which the first electrode **31** and the second electrode **32** are disposed. According to the present embodiment, heat can be dissipated to the substrate **10** via the insulating member **20** in such an inter-electrode region. The heat generated by the semiconductor layered body **33** can be efficiently transferred to the substrate **10** by dispersing the heat via a path passing through the first electrode **31**, the second electrode **32**, and the insulating member **20**, and the heat can be made less likely to be concentrated in the inter-electrode region of the semiconductor layered body **33**. Accordingly, the temperature of the active layer (junction temperature) can be reduced, and the light emission efficiency and reliability can be improved. Further, the improved reliability allows increased input power and improved optical output.

[0047] The first bonding member **41** preferably bonds the inner lateral surface of the first electrode **31** and the upper part of the first upper lateral surface **21A** of the insulating member **20**, and bonds the inner lateral surface of the second electrode **32** and the upper part of the second upper lateral surface **21B** of the insulating member **20**. Accordingly, heat can be easily transferred from the insulating member **20** to the first electrode **31** and the second electrode **32** via the first bonding member **41**.

[0048] The upper surface **21C** of the upper portion **21** of the insulating member **20** can be in contact with the first element surface **33A** of the semiconductor layered body **33** directly or via an adhesive layer having thermal conductivity. Accordingly, the heat generated by the semiconductor layered body **33** can be easily transferred to the insulating member **20**. As the adhesive layer having thermal conductivity, for example, a resin layer in which ceramics are dispersed, or a ceramic adhesive layer can be used.

[0049] The substrate **10** can further include a second bonding member **42**. The second bonding member **42** bonds the inner lateral surface **11C** of the first metal member **11** and the first lower lateral surface **22A** of the insulating member **20**, and bonds the inner lateral surface **12C** of the second metal member **12** and the second lower lateral surface **22B** of the insulating member **20**. Accordingly, heat can be easily transferred from the insulating member **20** to the first metal member **11** and the second metal member **12** via the second bonding member **42**.

[0050] The second bonding member **42** contains, for example, at least one metal selected from copper (Cu), chromium (Cr), nickel (Ni), silver (Ag), aluminum (Al), zinc (Zn), tin (Sn), titanium (Ti), cerium (Ce), zirconium (Zr), and magnesium (Mg). The second bonding member **42** can

contain a compound of the above-described metal or an alloy of the above-described metal.

[0051] The lower portion **22** of the insulating member **20** further has an upper surface **22C**. The upper surface **22C** joins the first upper lateral surface **21A** with the first lower lateral surface **22A**, and joins the second upper lateral surface **21B** with the second lower lateral surface **22B**. The upper surface **22C** joining the first upper lateral surface **21A** with the first lower lateral surface **22A** faces the first electrode **31** in the third direction **Z**. The upper surface **22C** joining the second upper lateral surface **21B** with the second lower lateral surface **22B** faces the second electrode **32** in the third direction **Z**.

[0052] The second bonding member **42** can be further located at the upper surface **22C** of the lower portion **22** of the insulating member **20**, a lower part of the first upper lateral surface **21A**, and a lower part of the second upper lateral surface **21B**. The second bonding member **42** is disposed continuously at the lower part of the first upper lateral surface **21A**, the upper surface **22C**, and the first lower lateral surface **22A**, and is disposed continuously at the lower part of the second upper lateral surface **21B**, the upper surface **22C**, and the second lower lateral surface **22B**. Accordingly, heat can be more easily transferred from the insulating member **20** to the first metal member **11** and the second metal member **12** via the second bonding member **42**.

[0053] The second bonding member **42** and the first bonding member **41** are disposed between the upper surface **22C** of the lower portion **22** of the insulating member **20** and each of the first electrode **31** and the second electrode **32**. Further, when the first metal member **11** includes the plating **13**, the plating **13** is disposed between the second bonding member **42** and the first bonding member **41**. Between the upper surface **22C** of the lower portion **22** of the insulating member **20** and the first electrode **31**, the first bonding member **41** is in contact with the first electrode **31** and the second electrode **32**, the first bonding member **41** is in direct contact with the second bonding member **42** via the plating **13**, and the second bonding member **42** is in contact with the upper surface **22C** of the lower portion **22** of the insulating member **20**. Accordingly, heat transferred from the semiconductor layered body **33** to the first electrode **31** and the second electrode **32** can also be easily dissipated to the insulating member **20**.

[0054] The upper surface **21C** of the upper portion **21** of the insulating member **20** is exposed from the second bonding member **42**. Accordingly, the first bonding member **41** is less likely to wet and spread on the upper surface **21C**, and thus a short-circuit between the first electrode **31** and the second electrode **32** via the first bonding member **41** can be made less likely to occur.

[0055] At the lower surface of the substrate **10**, the lower surface **22D** of the lower portion **22** of the insulating member **20** is exposed from the second bonding member **42**. Accordingly, in bonding the light source **1** to the mounting substrate, the bonding member is less likely to wet and spread on the lower surface **22D**, and thus a short-circuit between the first metal member **11** and the second metal member **12** via the bonding member can be made less likely to occur. Further, the width, in the first direction **X**, of the lower surface **22D** of the lower portion **22** is greater than the width, in the first direction **X**, of the upper surface **21C** of the upper portion **21**. Accordingly, the short-circuit between the first metal member **11** and the second metal member **12** can be made further less likely to occur.

[0056] The insulating member **20** further includes a metal compound layer **18** at the first lower lateral surface **22A**, the second lower lateral surface **22B**, the upper surface **22C** of the lower portion **22**, the first upper lateral surface **21A**, and the second upper lateral surface **21B**. The adhesion between the insulating member **20** and the first bonding member **41** and the adhesion between the insulating member **20** and the second bonding member **42** can be increased by the metal compound layer **18**, and gaps between the insulating member **20** and the first bonding member **41** and between the insulating member **20** and the second bonding member **42** can be made less likely to be formed. Accordingly, heat can be easily transferred between the insulating member **20** and the first electrode **31**, between the insulating member **20** and the second electrode **32**, between the insulating member **20** and the first metal member **11**, and between the insulating

member **20** and the second metal member **12**.

[0057] When the insulating member **20** is a nitride, the metal compound layer **18** can be formed into a metal nitride layer by a thermal reaction between the insulating member **20** and a metal contained in the second bonding member **42**, which will be described in a manufacturing method to be described below. For example, the insulating member **20** contains silicon nitride, the second bonding member **42** contains titanium or a titanium compound, and the metal compound layer **18** is a titanium nitride layer.

[0058] Alternatively, when the insulating member **20** is an oxide, the metal compound layer **18** can be a metal oxide layer.

[0059] Alternatively, when the insulating member **20** is a carbide, the metal compound layer **18** can be a metal carbide layer.

[0060] The light source **1** can include a light-transmissive member **34** on the upper surface of the light-emitting device **30** (the second element surface **33B** of the semiconductor layered body **33**, or the upper surface of the element substrate). The light-transmissive member **34** and the upper surface of the light-emitting device **30** are bonded to each other via a light-transmissive resin layer. Alternatively, the light-transmissive member **34** and the light-emitting surface of light-emitting device **30** can be directly bonded to each other.

[0061] As the light-transmissive member **34**, for example, a wavelength conversion member, a glass member, a ceramic member, or the like can be used. The light-transmissive member **34** can have a flat-plate shape or a lens shape, and can have an uneven shape on at least one surface. The wavelength conversion member converts the wavelength of at least a part of light emitted by the active layer of the semiconductor layered body **33**. As the wavelength conversion member, a sintered compact can be used that is formed by sintering powder of a phosphor so as to be substantially made of only the phosphor. Alternatively, as the wavelength conversion member, a light-transmissive material containing a phosphor can be used. Examples of the light-transmissive material that can be used include a ceramic, a resin, and a glass. Examples of the phosphor that can be used include a cerium-activated yttrium-aluminum-garnet-based phosphor (for example, $(Y,Gd).sub.3(Al,Ga).sub.5O.sub.12:Ce$), and a cerium-activated lutetium-aluminum-garnet-based phosphor (for example, $Lu.sub.3(Al,Ga).sub.5O.sub.12:Ce$).

[0062] The light source **1** can further include a covering member **50**. The covering member **50** can have reflectivity with respect to the light emitted by the active layer. For example, the covering member **50** includes a light-transmissive base material and light-reflective particles dispersed in the base material. As the material of the base material, a resin, glass, or a ceramic can be used. Examples of the light-reflective particles include titanium oxide, silicon oxide, zinc oxide, magnesium oxide, magnesium carbonate, magnesium hydroxide, calcium carbonate, calcium hydroxide, calcium silicate, magnesium silicate, barium titanate, barium sulfate, aluminum hydroxide, aluminum oxide, and zirconium oxide. When the light-emitting device **30** is an LD, the covering member **50** need not necessarily be provided.

[0063] The outer lateral surface **11D** of the first metal member **11** and the outer lateral surface **12D** of the second metal member **12** are preferably exposed from the covering member **50**. A bonding member, which is used in bonding the light source **1** to the mounting substrate, can wet and rise up the exposed outer lateral surfaces **11D** and **12D**. Accordingly, heat generated by the light-emitting device **30** can be dissipated to the mounting substrate from the outer lateral surface **11D** of the first metal member **11** via the bonding member, and from the outer lateral surface **12D** of the second metal member **12** via the bonding member.

[0064] As illustrated in FIG. 3, a portion **11E** of the first metal member **11** can be located between the upper surface **22C** of the lower portion **22** of the insulating member **20** and the first electrode **31**, and a portion **12E** of the second metal member **12** can be located between the upper surface **22C** of the lower portion **22** of the insulating member **20** and the second electrode **32**. Accordingly, an area in which the first electrode **31** and the first metal member **11** face each other and an area in

which the second electrode **32** and the second metal member **12** face each other can be increased, and thus heat dissipation from the first electrode **31** and the second electrode **32** to the first metal member **11** and the second metal member **12** can be improved.

Manufacturing Method of Substrate

[0065] Next, an example of a method for manufacturing the substrate **10** will be described with reference to FIGS. **4** to **11**.

[0066] As illustrated in FIGS. **4** and **5**, the method for manufacturing the substrate **10** includes a step of preparing a ceramic sintered compact **100**. FIG. **5** is a schematic cross-sectional view taken along a line V-V in FIG. **4**.

[0067] The ceramic sintered compact **100** includes a plurality of the insulating members **20** extending in the second direction Y and arranged side by side so as to be spaced apart from each other in the first direction X. A through hole portion **60** extending in the second direction Y is located between the insulating members **20** adjacent to each other in the first direction X. As illustrated in FIG. **5**, the through hole portion **60** includes a first hole portion **61** adjacent to the upper portion **21** of the insulating member **20**, and a second hole portion **62** adjacent to the lower portion **22** of the insulating member **20**. The first hole portion **61** and the second hole portion **62** are connected to each other in the third direction Z. The width of the first hole portion **61** in the first direction X is greater than the width of the second hole portion **62** in the first direction X. For example, end portions of the plurality of insulating members **20** in the second direction Y are integrally connected to a coupling member extending in the first direction X.

[0068] The step of preparing the ceramic sintered compact **100** can include a step of forming the through hole portion **60** in a ceramic plate, which is a sintered compact subjected to a baking treatment, using a method such as laser processing or blade processing. By forming the through hole portion **60** in the ceramic plate, which is the sintered compact, using the above-described processing method, it is possible to form the plurality of insulating members **20** in which variations in the shape are reduced.

[0069] As illustrated in FIG. **6**, the method of manufacturing the substrate **10** includes a step of filling the through hole portion **60** with a metal paste **110**.

[0070] The metal paste **110** contains, for example, at least one metal selected from copper (Cu), chromium (Cr), nickel (Ni), silver (Ag), aluminum (Al), zinc (Zn), tin (Sn), titanium (Ti), cerium (Ce), zirconium (Zr), and magnesium (Mg). Further, the metal paste **110** can contain a compound of the above metal or an alloy of the above metal. Further, the metal paste **110** contains active metal powder. The active metal powder can be at least one selected from $\text{TiH}_{0.2}$, $\text{CeH}_{0.2}$, $\text{ZrH}_{0.2}$, and $\text{MgH}_{0.2}$. Further, the metal paste **110** can contain an organic solvent. The viscosity of the metal paste **110** can be adjusted in accordance with the type and amount of the organic solvent. As the organic solvent, for example, a resin such as acrylic, epoxy, urethane, ethyl cellulose, silicone, phenol, polyimide, polyurethane, melamine, or urea can be used. The metal paste **110** can contain an inorganic filler. When the metal paste **110** contains the inorganic filler, volumetric shrinkage in a baking treatment described below can be reduced. As the inorganic filler, for example, aluminum nitride (AlN), silicon nitride ($\text{Si}_{0.3}\text{N}_{0.4}$), aluminum oxide ($\text{Al}_{0.2}\text{O}_{0.3}$), or the like can be used. As an example, the metal paste **110** preferably contains an silver-copper alloy with a mixing ratio of silver (Ag) and copper (Cu) of 72:28 in a range from 60 wt. % to 99 wt. %, titanium hydroxide ($\text{TiH}_{0.2}$) as active metal powder in a range from 0.5 wt. % to 10 wt. %, an organic binder in a range from 0.1 wt. % to 10 wt. %, and an inorganic filler in a range from 1 wt. % to 20 wt. %.

[0071] As illustrated in FIG. **7**, the method of manufacturing the substrate **10** includes a step of disposing a metal body **120** in the metal paste **110**. The metal body **120** extends in the second direction Y. A plurality of the metal bodies **120** can be prepared from a metal plate, for example, using a method such as punching, laser machining, etching, or the like. The cross-sectional shape, the upper surface shape, and the lower surface shape of the metal body **120** are rectangular or

square. Accordingly, the plurality of metal bodies **120**, in which variations in the shape are reduced, can be prepared from the metal plate using the above-described method. In a step described below, the metal body **120** is diced along the first direction X and the second direction Y in a plan view, so as to be separated into the above-described first metal member **11** and second metal member **12**.

[0072] The method of manufacturing the substrate **10** can include, after disposing the metal body **120** in the metal paste **110**, as necessary, a step of further forming the metal paste **110** so as to cover the upper surface of the metal body **120**, the lower surface of the metal body **120**, the upper surface **21C** of the upper portion **21** of the insulating member **20**, and the lower surface **22D** of the lower portion **22** of the insulating member **20**, as illustrated in FIG. **8**. Accordingly, volumetric shrinkage of a metal body (second bonding member **42**) described below can be reduced. The metal body is obtained by performing a baking treatment on the metal paste **110**.

[0073] The method of manufacturing the substrate **10** includes, after disposing the metal body **120** in the metal paste **110**, a step of performing a baking treatment in which the metal paste **110** is baked at a temperature in a range from 780° C. to 850° C., for example. By the baking treatment, the organic solvent in the metal paste **110** is evaporated, and further, the powder of the metal or the alloy is melted to form the second bonding member **42**, which is a continuous metal body.

[0074] For example, when the metal paste **110** contains silver-copper alloy powder as the main component, the melting point of the silver-copper alloy powder is 780° C., which is lower than the melting point (1084° C.) of copper powder and the melting point (962° C.) of silver powder, and thus the baking treatment can be performed at a lower temperature than that when the metal paste **110** contains the copper powder or the silver powder as the main component.

[0075] Further, by the baking treatment, the metal contained in the metal paste **110** and the insulating member **20** thermally react with each other, and the metal compound layer **18** is formed on the surfaces of the insulating member **20**, which have been in contact with the metal paste **110** (the first upper lateral surface **21A**, the second upper lateral surface **21B**, the first lower lateral surface **22A**, the second lower lateral surface **22B**, and the upper surface **22C** of the lower portion **22**). The metal compound layer **18** increases the bonding force between the insulating member **20** and the second bonding member **42**. For example, when the insulating member **20** contains silicon nitride, and the metal paste **110** contains titanium hydride as the active metal powder, a titanium nitride layer is formed as the metal compound layer **18**.

[0076] After the baking treatment, the second bonding member **42**, which covers the upper surface of the metal body **120**, the lower surface of the metal body **120**, the upper surface **21C** of the upper portion **21** of the insulating member **20**, and the lower surface **22D** of the lower portion **22** of the insulating member **20**, is removed by a polishing treatment. Accordingly, as illustrated in FIG. **9**, the upper surface of the metal body **120**, the lower surface of the metal body **120**, the upper surface **21C** of the upper portion **21** of the insulating member **20**, and the lower surface **22D** of the lower portion **22** of the insulating member **20** are exposed from the second bonding member **42**. The metal body **120** is bonded, by the second bonding member **42**, to the first upper lateral surface **21A**, the second upper lateral surface **21B**, the first lower lateral surface **22A**, and the second lower lateral surface **22B** of the insulating member **20**.

[0077] The method for manufacturing the substrate **10** includes, after the above-described baking treatment, a step of removing a part of the upper surface side of the metal body **120** and a part of the upper surface side of the second bonding member **42** to expose a part of the first upper lateral surface **21A** and a part of the second upper lateral surface **21B** of the insulating member **20** from the second bonding member **42**, as illustrated in FIG. **10**. For example, the metal body **120** and the second bonding member **42** can be removed using a method such as etching or blasting.

[0078] The metal compound layer **18** is not removed and remains at the first upper lateral surface **21A** and the second upper lateral surface **21B** exposed from the second bonding member **42**.

Accordingly, the above-described first bonding member **41** easily wets and spreads on the metal compound layer **18**.

[0079] The method of manufacturing the substrate **10** can include a step of forming a plating **131** and a plating **132**, as illustrated in FIG. **11**, as necessary. The plating **131** and the plating **132** can be formed by electrolytic plating or electroless plating. The plating **131** is formed at the upper surface of the metal body **120**, and the plating **132** is formed at the lower surface of the metal body **120**. At a time of dicing the metal body **120**, the plating **131** is separated into the above-described plating **13** and plating **14**, and the plating **132** is separated into the above-described plating **15** and plating **16**.

Method for Bonding Light-Emitting Device to Substrate

[0080] The first bonding member **41** is supplied to the upper surface of the metal body **120** of the substrate **10** obtained by the steps described above, and the light-emitting device **30** is disposed on the substrate **10** such that the upper portion **21** of the insulating member **20** is located between the first electrode **31** and the second electrode **32** of the light-emitting device **30**. Each of the first electrode **31** and the second electrode **32** is disposed on the first bonding member **41**. There are gaps between the first upper lateral surface **21A** of the insulating member **20** and the inner lateral surface of the first electrode **31** and between the second upper lateral surface **21B** of the insulating member **20** and the inner lateral surface of the second electrode **32**. In this state, a reflow treatment is performed in which the first bonding member **41** is heated and melted.

[0081] The melted first bonding member **41** wets and spreads between the first electrode **31** and the metal body **120** and between the second electrode **32** and the metal body **120**, and further wets and rises up between the first upper lateral surface **21A** of the insulating member **20** and the inner lateral surface of the first electrode **31** and between the second upper lateral surface **21B** of the insulating member **20** and the inner lateral surface of the second electrode **32**. When the first bonding member **41** is cured after the reflow treatment, the first electrode **31** and the metal body **120** are bonded to each other, the second electrode **32** and the metal body **120** are bonded to each other, the first electrode **31** and the upper portion **21** of the insulating member **20** are bonded to each other, and the second electrode **32** and the upper portion **21** of the insulating member **20** are bonded to each other.

[0082] Since the upper portion **21** of the insulating member **20** is located between the first electrode **31** and the second electrode **32** of the light-emitting device **30**, it is possible to restrict rotation of the light-emitting device **30** in the XY plane and a large positional shift of the light-emitting device **30** in the first direction X at a time of bonding during which the first bonding member **41** is melted and has fluidity.

[0083] After the light-emitting device **30** is bonded to the substrate **10**, outer lateral surfaces of the light-emitting device **30** are covered with the covering member **50**, as necessary. Thereafter, the metal body **120** is diced along the first direction X and the second direction Y, and the insulating member **20** is also cut at the dicing positions of the metal body **120** along the first direction X. As a result, a plurality of the singulated light sources **1** are obtained.

[0084] The embodiments of the present invention have been described above with reference to specific examples. However, the present invention is not limited to these specific examples. All aspects that can be practiced by a person skilled in the art changing the design as appropriate based on the above-described embodiments of the present invention are also included in the scope of the present invention, as long as they encompass the spirit of the present invention. In addition, in the spirit of the present invention, a person skilled in the art can conceive of various variations and modifications, and those variations and modifications will also fall within the scope of the present invention.

Claims

1. A light source comprising: a substrate; a light-emitting device disposed on the substrate; and a first bonding member disposed between the substrate and the light-emitting device, wherein the

substrate comprises a first metal member having a first upper surface and a first lower surface located on a side opposite to the first upper surface, a second metal member having a second upper surface and a second lower surface located on a side opposite to the second upper surface, the second metal member being spaced apart from the first metal member in a first direction, and an insulating member comprising a lower portion located between the first metal member and the second metal member in the first direction, and an upper portion extending, from the lower portion, to be higher than the first upper surface and the second upper surface, the light-emitting device comprises a semiconductor layered body having a first element surface facing the substrate and a second element surface located on a side opposite to the first element surface, a first electrode disposed at the first element surface, and a second electrode disposed at the first element surface to be spaced apart from the first electrode in the first direction, the first bonding member is disposed between the first electrode and the first upper surface of the first metal member and bonds the first electrode and the first metal member, and is disposed between the second electrode and the second upper surface of the second metal member and bonds the second electrode and the second metal member, the upper portion of the insulating member is located between the first electrode and the second electrode in the first direction, and the lower portion of the insulating member is exposed.

2. The light source according to claim 1, wherein the upper portion of the insulating member has a first upper lateral surface facing an inner lateral surface of the first electrode, the upper portion of the insulating member has a second upper lateral surface facing an inner lateral surface of the second electrode, the first bonding member bonds the inner lateral surface of the first electrode and an upper part of the first upper lateral surface of the insulating member, and the first bonding member bonds the inner lateral surface of the second electrode and an upper part of the second upper lateral surface of the insulating member.

3. The light source according to claim 2, wherein the lower portion of the insulating member has a first lower lateral surface facing an inner lateral surface of the first metal member, the lower portion of the insulating member has a second lower lateral surface facing an inner lateral surface of the second metal member, the substrate further comprises a second bonding member bonding the inner lateral surface of the first metal member and the first lower lateral surface, and the second bonding member bonding the inner lateral surface of the second metal member and the second lower lateral surface.

4. The light source according to claim 3, wherein the second bonding member is further provided at an upper surface of the lower portion of the insulating member, a lower part of the first upper lateral surface, and a lower part of the second upper lateral surface.

5. The light source according to claim 1, wherein a width, in the first direction, of the lower portion of the insulating member is greater than a width, in the first direction, of the upper portion of the insulating member.

6. The light source according to claim 3, wherein a width, in the first direction, of the lower portion of the insulating member is greater than a width, in the first direction, of the upper portion of the insulating member, and at a lower surface of the substrate, a width, in the first direction, of the lower portion of the insulating member exposed from the second bonding member is greater than a width, in the first direction, of the upper portion of the insulating member exposed from the second bonding member.

7. The light source according to claim 1, wherein an upper surface of the upper portion of the insulating member is in contact with the first element surface of the light-emitting device.

8. The light source according to claim 1, wherein the insulating member is made of a ceramic.

9. The light source according to claim 4, wherein the insulating member further comprises a metal compound layer at the first lower lateral surface, the second lower lateral surface, the upper surface of the lower portion of the insulating member, the first upper lateral surface, and the second upper lateral surface.

10. The light source according to claim 9, wherein the insulating member contains silicon nitride,

the second bonding member contains titanium or a titanium compound, and the metal compound layer is a titanium nitride layer.

11. The light source according to claim 1, wherein the insulating member is formed by covering a metal with a ceramic, glass, or a resin.

12. The light source according to claim 1, wherein the first lower surface of the first metal member and the second lower surface of the second metal member are exposed.

13. The light source according to claim 1, wherein the first metal member includes a first plating, a lower surface of the first plating forms the first lower surface of the first metal member, the second metal member includes a second plating, a lower surface of the second plating forms the second lower surface of the second metal member, and the lower surface of the first plating and the lower surface of the second plating are exposed.

14. A substrate comprising: a first metal member having a first upper surface and a first lower surface located on a side opposite to the first upper surface; a second metal member having a second upper surface and a second lower surface located on a side opposite to the second upper surface, the second metal member being spaced apart from the first metal member in a first direction; and an insulating member comprising a lower portion located between the first metal member and the second metal member in the first direction, and an upper portion extending, from the lower portion, to be higher than the first upper surface and the second upper surface, wherein a width, in the first direction, of the lower portion of the insulating member is greater than a width, in the first direction, of the upper portion of the insulating member, the lower portion of the insulating member has a first lower lateral surface facing an inner lateral surface of the first metal member, the lower portion of the insulating member has a second lower lateral surface facing an inner lateral surface of the second metal member, the substrate further comprises a bonding member bonding the inner lateral surface of the first metal member and the first lower lateral surface, the bonding member bonding the inner lateral surface of the second metal member and the second lower lateral surface, and the bonding member being located at an upper surface of the lower portion of the insulating member.

15. The substrate according to claim 14, wherein the insulating member is made of a ceramic, and the substrate further comprises a metal compound layer provided at an interface between the insulating member and the bonding member.
